

Editorial: Problem A - Ship Wars

Setter: Hudson

Problem Overview

We are given N entries, each consisting of a fandom name (“Bocchi” or “OshiNoKo”) and an associated score. We need to calculate the total score for each fandom and determine which one is strictly greater, or if they are tied.

Solution

This is a straightforward implementation problem. We need to maintain two integer variables, say `total_bocchi` and `total_onk`, both initialized to 0.

We iterate through the input N times. In each iteration, we read a string F and an integer S .

- If the string F is “Bocchi”, we add S to `total_bocchi`.
- Otherwise (if F is “OshiNoKo”), we add S to `total_onk`.

After processing all inputs, we compare the two totals:

- If `total_bocchi > total_onk`, print “Bocchi”.
- If `total_onk > total_bocchi`, print “OshiNoKo”.
- If they are equal, print “Tie”.

Complexity Analysis

- **Time Complexity:** $O(N)$ because we iterate through the list of ships exactly once.
- **Space Complexity:** $O(1)$ (or $O(L)$ where L is the max string length) as we only store a few variables and the current input string.

Implementation (C++)

```
1 #include <iostream>
2 #include <string>
3
4 using namespace std;
5
6 int main() {
7     int N;
8     if (!(cin >> N)) return 0;
9 }
```

```

10     long long bocchi_score = 0;
11     long long onk_score = 0;
12
13     for (int i = 0; i < N; ++i) {
14         string fandom;
15         int score;
16         cin >> fandom >> score;
17
18         if (fandom == "Bocchi") {
19             bocchi_score += score;
20         } else {
21             onk_score += score;
22         }
23     }
24
25     if (bocchi_score > onk_score) {
26         cout << "Bocchi" << endl;
27     } else if (onk_score > bocchi_score) {
28         cout << "OshiNoKo" << endl;
29     } else {
30         cout << "Tie" << endl;
31     }
32
33     return 0;
34 }

```